MASTER OF BUSINESS ADMINISTRATION

SEMESTER 3

DADS304

VISUALIZATION

Unit 9

Other Features

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1. INTRODUCTION

Tableau Groups are sets of multiple members combined into a single dimension for the purpose of creating a higher level dimension. Grouping single-dimensional members in Tableau automatically creates a new dimension with the group name at the end. The original dimension of the members is not altered by Tableau. Group is used to combine members present in a field.

1.1 Learning Objectives

After studying this unit, you will be able to:

- *Explain features of Tableau*
- *Explain Groups and calculated fields*
- *Steps to calculate sets and calculated fields*

2. GROUPS

A group is used to combine members in a field. Using group, you can aggregate the values of 'Furniture' and 'Office Supplies'. Using Tableau, aggregated values of 'Furniture' and 'Office Supplies' can be shown in visuals after grouping the data. Following is a procedure for grouping data in Tableau.

2.1 Steps to Calculate Groups in TABLEAU

Step 1: Right-click on the dimension 'Category'. Click on 'Create' option. Select 'Group' option.

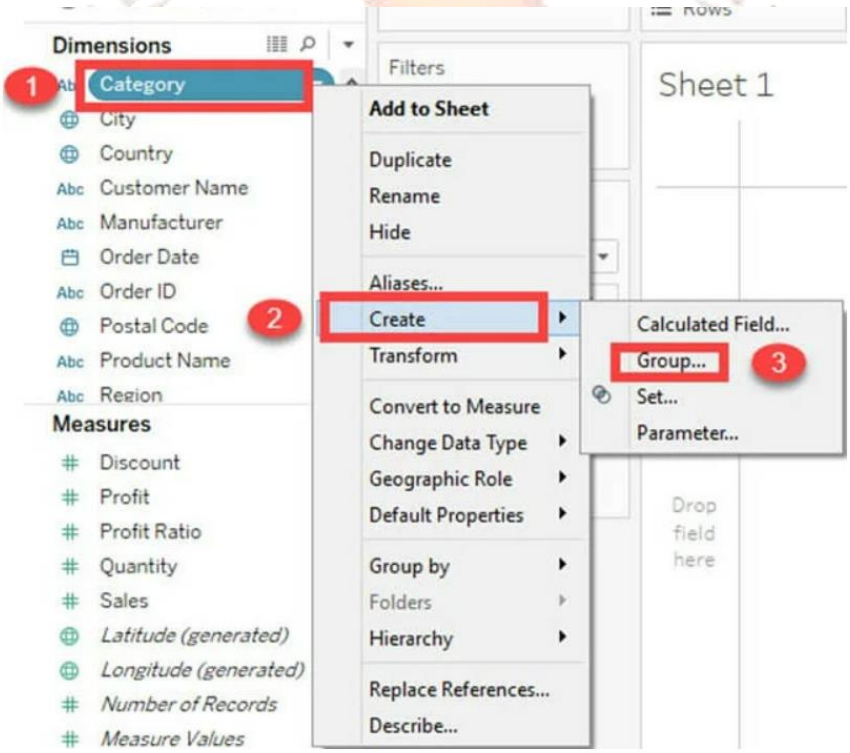


Fig 9.1 Creating groups in Tableau

Step 2: It opens the 'Create group' window. Type the name of the group data in Tableau. Select the members to be grouped. Click on 'Group 'button.

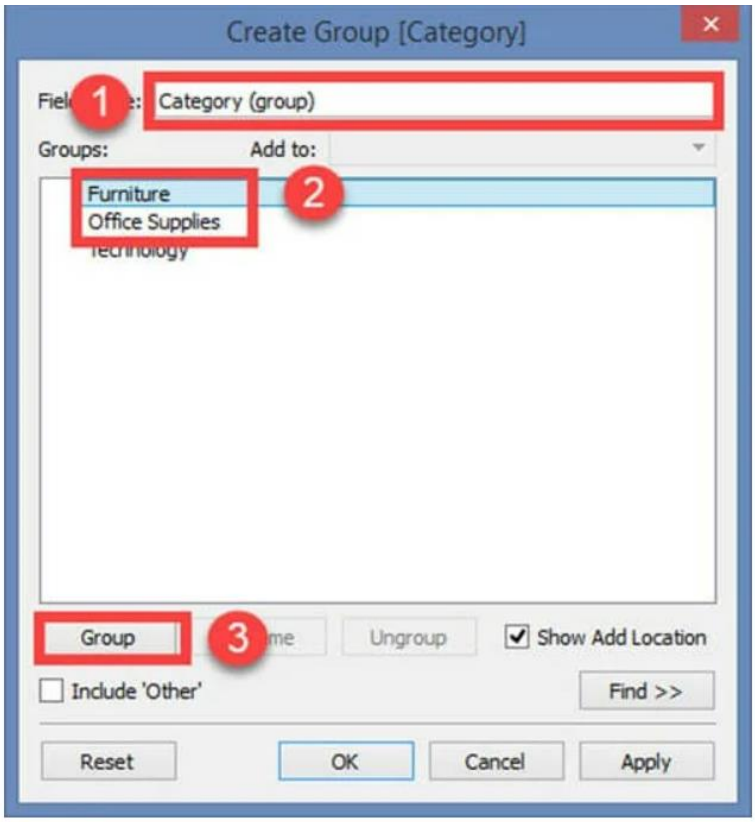


Fig 9.2 Creating groups in Tableau

Step 3: In Edit Group Window, It creates groups in Tableau of 'Furniture' and 'Office supplies'. Click on Ok to create the group.

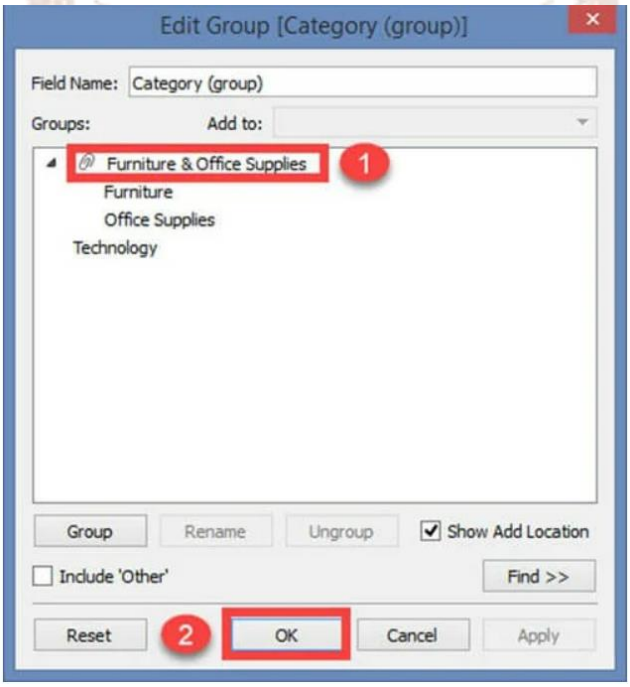


Fig 9.3 Creating groups in Tableau

Step 4: A group in Tableau with the name of Category (Group) and added in the dimension list is created. This can be used for visualizing the group by in Tableau method for members present in a field.

The following image explains the functionality of Tableau create group. The sum of sales is visualized for both furniture and office supplies for grouping in Tableau.

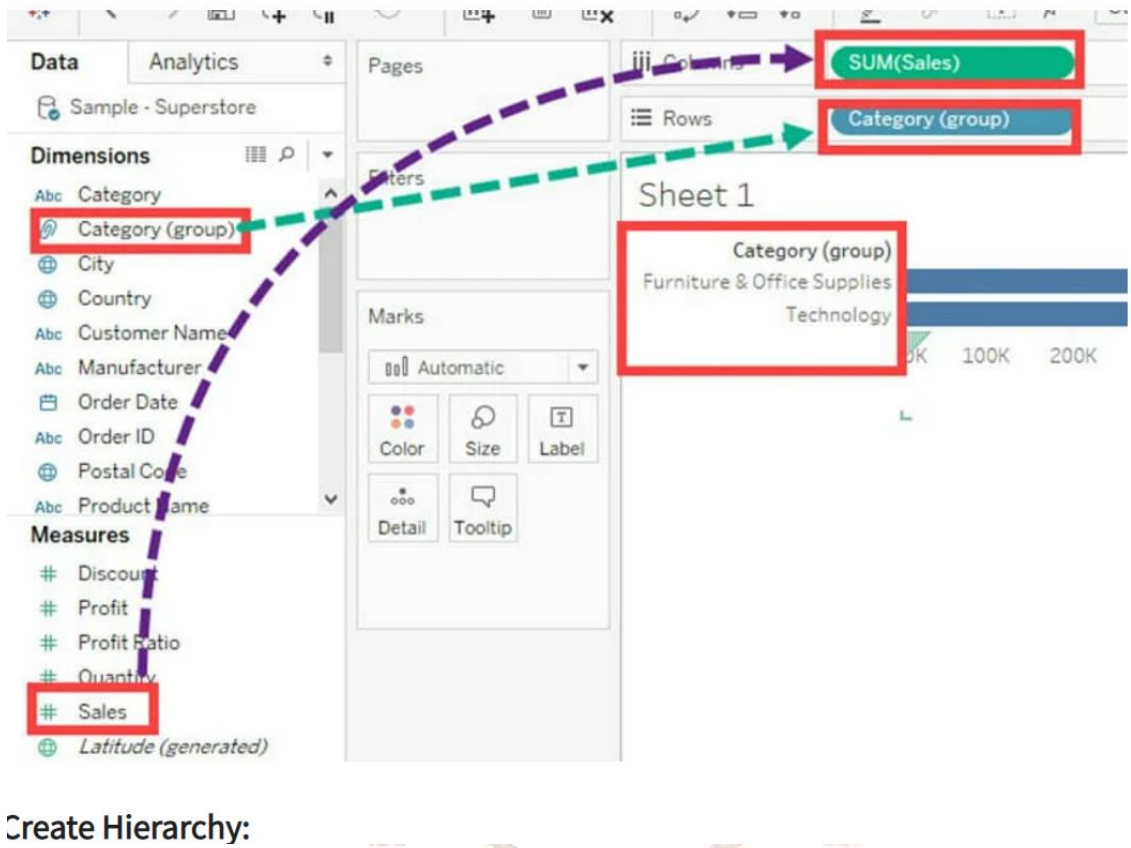


Fig 9.4 Creating groups in Tableau

SELF-ASSESSMENT QUESTIONS – 1

1. Data Values available for the visualization_____
2. _____is not a Trend Line model

Answers

1. Table calculation
2. Binomial trend line

3. CALCULATED FIELDS AND PARAMETERS

Software or programming languages use parameters as dynamic values to replace constant values in calculations, filters, or references. Consider a case where you want to generate a report each month that shows the number of employees earning more than 1,00,000/- INR per month.

Step 1: In the Data pane, right-click a field in which you want to create a parameter and then select Create -> Parameter.

Step 2: Give the field a name and provide an optional comment so as to explain your parameter

Step 3: Provide the type of data that it accepts

Step 4: Provide a current value, which will also be the default value of the parameter

Step 5: Provide a specific display format on the parameter control

Step 6: Based on the option selected for 'Allowable Values', you must provide the values to the parameter defined

Step 7: When all the above steps are completed, click OK to complete the process. The newly created parameter will be listed on the Parameters section (i.e., the bottom of the Data pane).

Using parameters in calculations is as simple as dragging them from the Data pane or dropping them on the Calculation editor (either replacing a particular part of the formula or at a new location). Click on the first parameter and click on Duplicate to create a new one with the same configuration as the first. By renaming it, it is possible to create two parameters with the same configuration. There are two parameters named Placeholder 1 Selector and Placeholder 2 Selector. On Analysis, select Create Calculated Field to open up the calculation editor. Click Ok to close the calculation editor. Now create a duplicate calculated field with the same configuration.

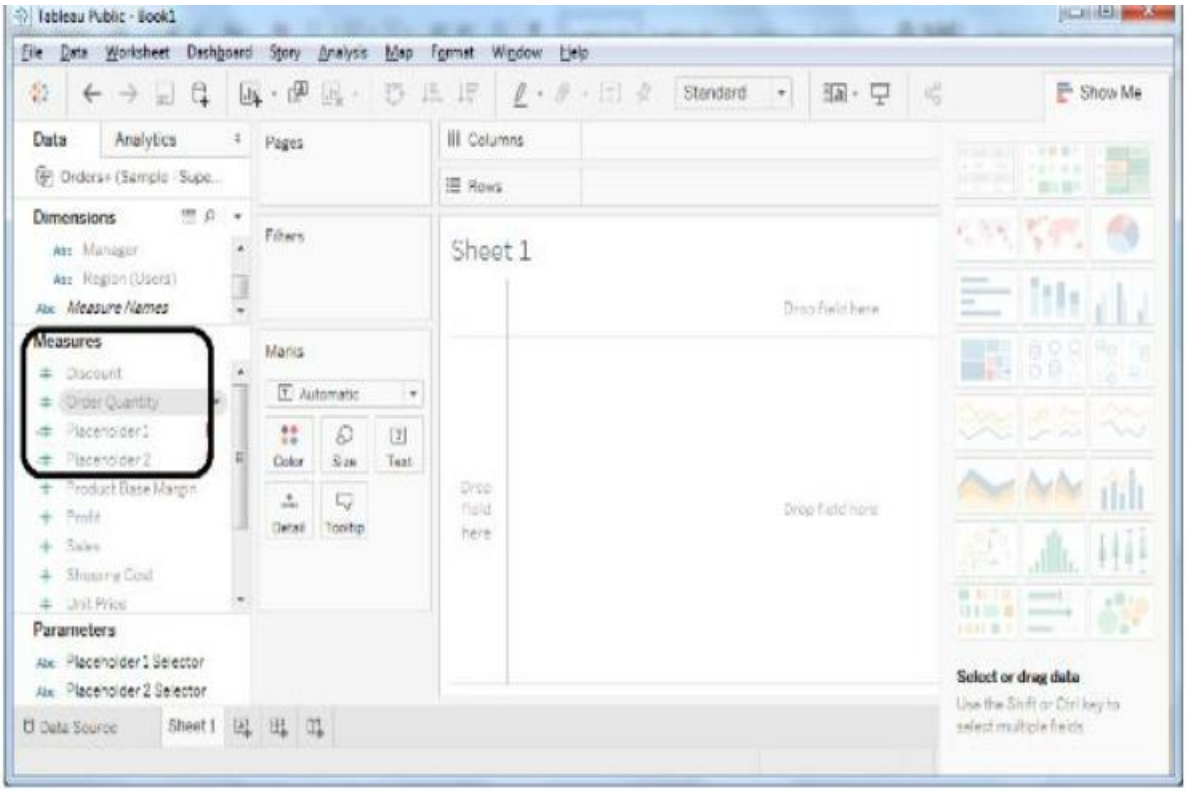


Fig 9.5 Creating calculated fields and parameters in Tableau

After the previous stages, it is simpler to set up the view aswe can simply drag and drop these computed fields onto the area of the tool.Drag placeholders two to the columns and one to the rows.Drag the Customer Name field to Detail and the Region field to Color.Click on each parameter in the Data Pane's parameters section, then select "Show Parameter Control." The other parameter should be treated similarly. The Tableau Desktop view is now ready . It is possible to select the data that will be displayed on your X and Y axes, respectively, using the parameter controls. The data can be altered by creating permutations and combinations based on the parameters provided.

SELF-ASSESSMENT QUESTIONS – 2

3. How do you find the field is continuous in tableau?
4. What percent of total profits do the top 10 customer by Sales represent?

Answers

3. Green colour
4. 5.03%

4. SETS

There are two types of sets: dynamic sets and fixed sets. The members of a dynamic set change when the underlying data changes. Dynamic sets can only be based on a single dimension.

9.4 STEPS INVOLVED IN CREATING DYNAMIC SETS

Step 1: Right-click a dimension in the Data window and choose Create > Set.

Step 2: Set up your set in the Create Set dialogue box. The following tabs can be used to configure your set:

Step 3: General: To choose one or more values that will be taken into account while calculating the set, use the General tab.

Step 4: Alternately, the Use all option to always take into account all members, regardless of whether new members are added or withdrawn can be used.

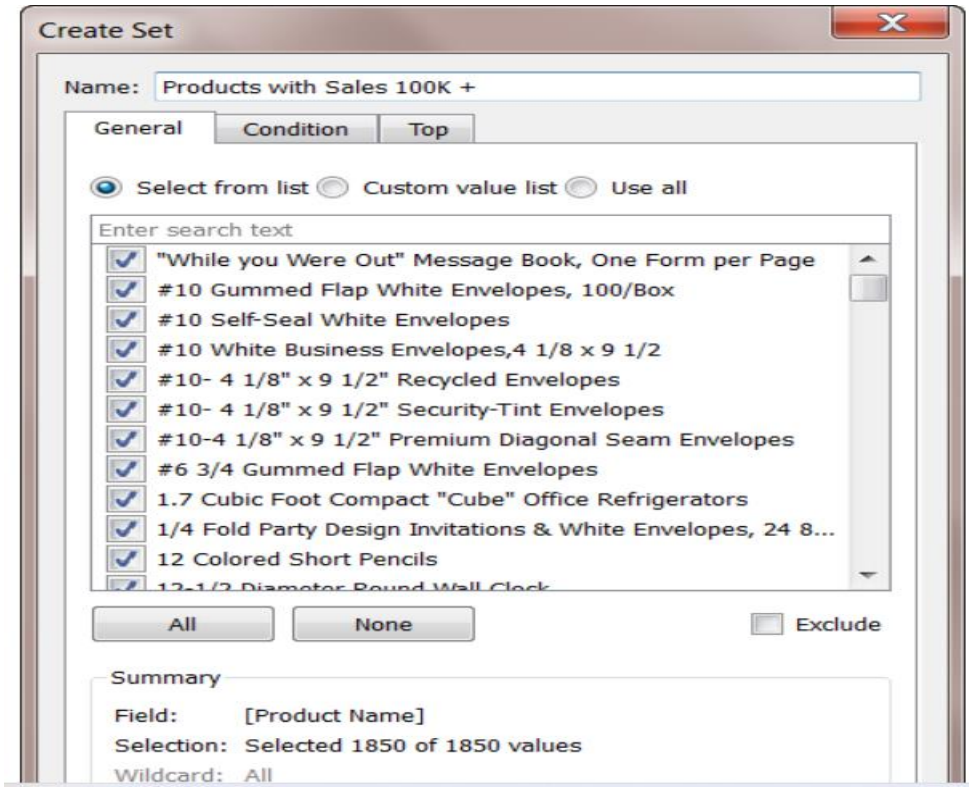


Fig 9.6 Creating dynamic sets

Step 5: Condition: To specify guidelines for choosing which members to include in the collection, use the Condition tab. For instance, establish a condition based on total sales that only takes into account goods with sales greater than $100,000.

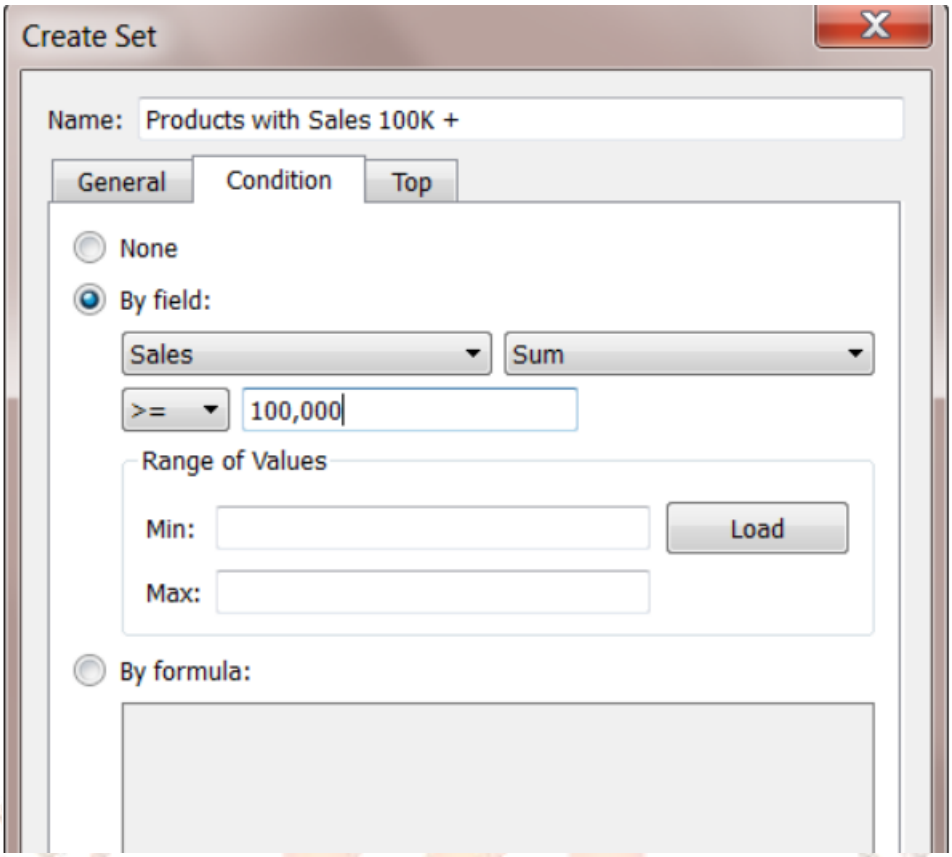


Fig 9.7 Creating dynamic sets

Step 6: Top: Use the Top tab to define limits on what members to include in the set. For example, specify a limit that is based on total sales that only includes the top 5 products based on their sales. When finished, click OK.

4.1 Steps Involved In Creating Fixed Sets

Step 1: In the visualization, select one or more marks (or headers) in the view. Right-click the mark(s) and select Create Set. In the Create Set dialog box, type a name for the set.

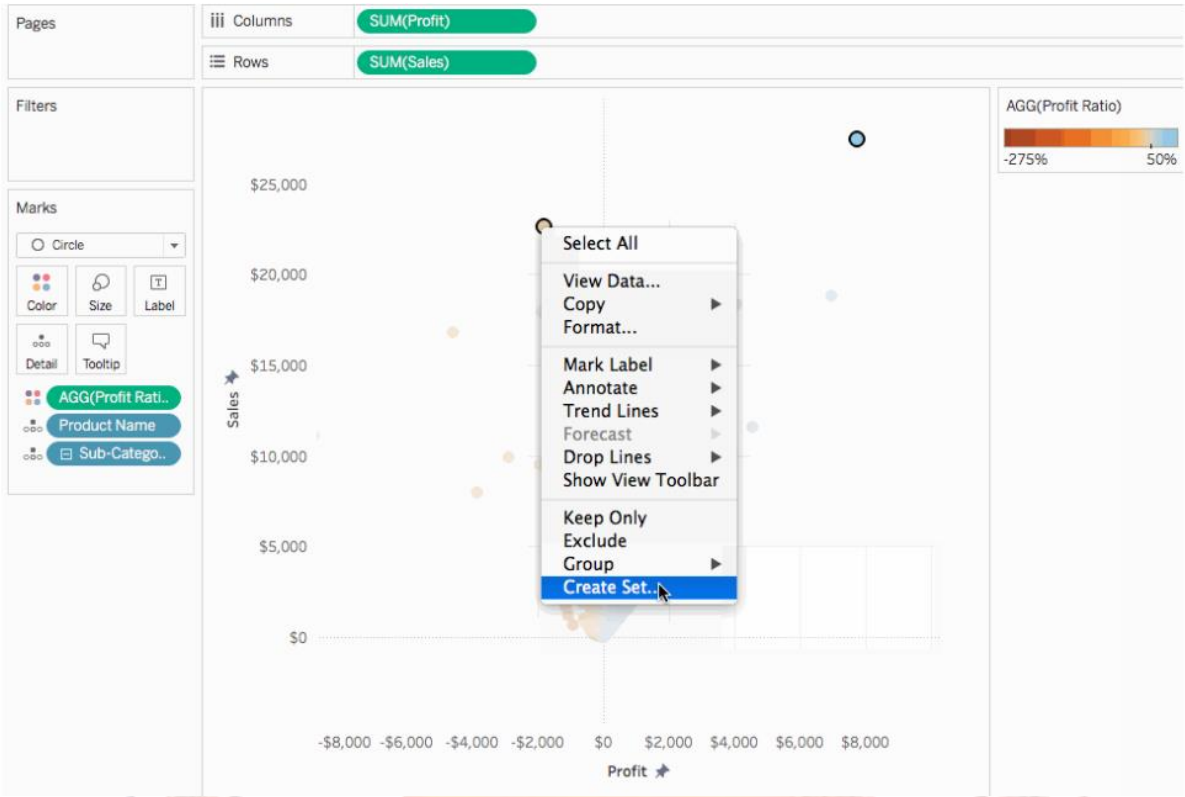


Fig 9.8 Creating fixed sets

Step 2: The members listed in the dialogue box are included in the set by default. Instead, choose to exclude certain members. The set will contain every member that are not selected when to exclude. By clicking the red "x" button that shows when you hover over a column heading, it is possible to eliminate any measurements that are not required to be taken into account. Click the red "x" button that appears when you hover over any individual rows that are not required to be part of the collection.

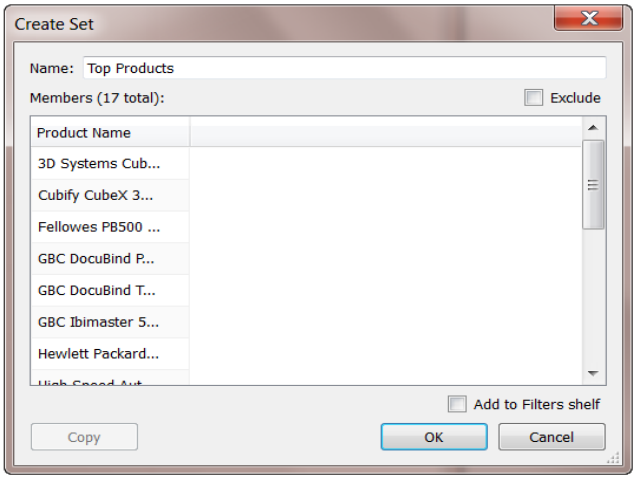


Fig 9.9 Creating fixed sets

Step 3: Finally click OK.

SELF-ASSESSMENT QUESTIONS – 3

5. Power BI is a product of_____
6. Who is the parent company of Tableau?

Answers

5. Microsoft
6. Sales Force

5. TRENDS

Trend lines are used to forecast whether a certain trend in a variable will continue. By observing the trend in both variables at once, it is also possible to determine the correlation between them. There are numerous mathematical techniques for drawing trend lines. Tableau offers four choices. Linear, Logarithmic, Exponential, and Polynomial are the types. To construct a Trend Line, Tableau needs a time dimension and a measure field.

Step 1: Drag the measure Sales to the Rows shelf and the dimension Order date to the Column shelf. Select a line chart as the chart type. Navigate to model Trend Line under the Analysis menu. The option to add several types of trend lines appears when the button for Trend Line is clicked. Select the linear model as displayed in the screenshot below.

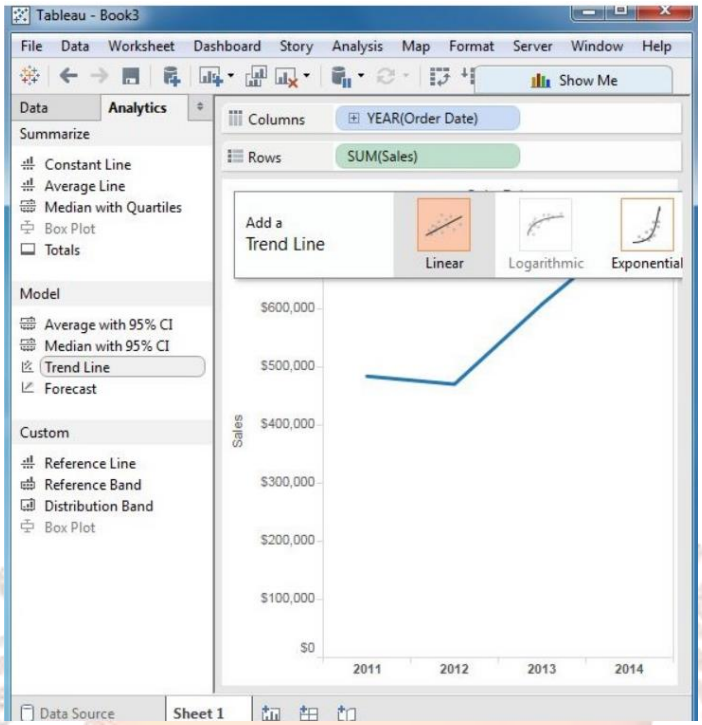


Fig 9.10 Creating trends

Step 2: A different trend lines after completing the aforementioned procedure is obtained. The P-Value and R-Squared values are also displayed, along with the mathematical equation for the correlation between the fields.

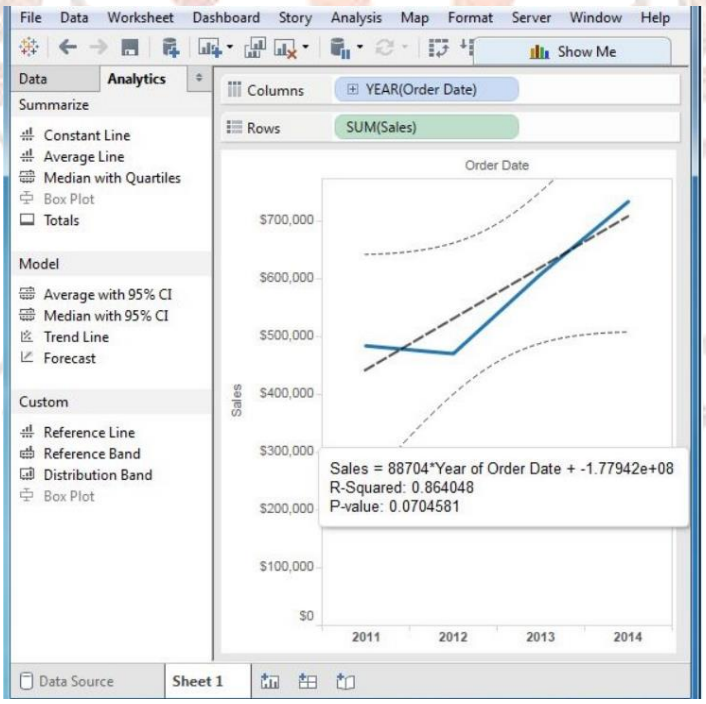


Fig 9.11 Creating trends

Step 3: Right-click on the chart and select the option Describe Trend Line to get a detailed description of the Trend Line chart. It shows the coefficients, intercept value, and the equation. These details can also be copied to the clipboard and used in further analysis.

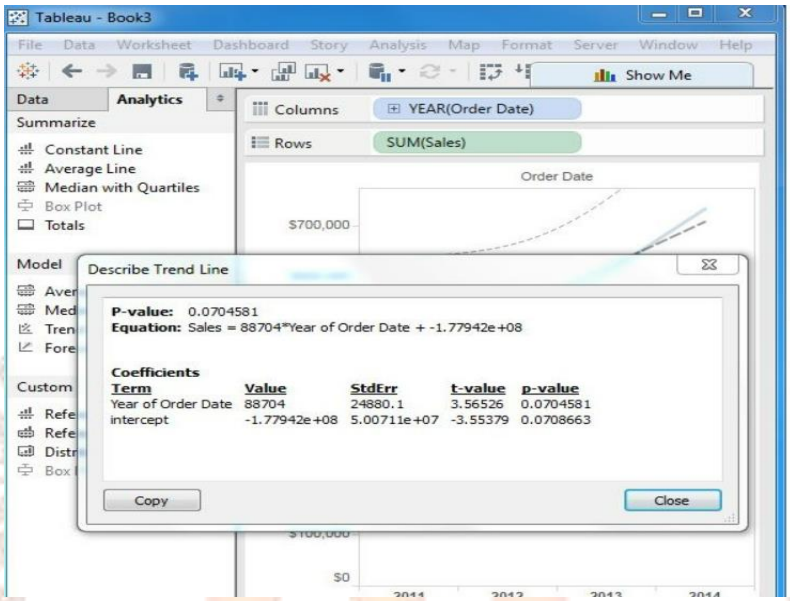


Fig 9.12 Creating trends

6. REFERENCE LINES AND FORECAST

A reference line in Tableau is simply a line that gets drawn on a chart that represents another measure or point of reference. Tableau Reference lines can be useful in providing context to the related chart. For example, a line showing the median will visually show the difference of each mark in the chart relative to the median.

Step 1: From Dimensions, drag Year to Rows.

Step 2: From Dimensions, drag Genre to Rows to the right of the Year pill.

Step 3: From Measures, drag Worldwide Gross Amount to Columns.

Step 4: Hover over the Worldwide Gross Amount axis, and click on the sort icon once to sort the bars in descending order.

Step 5: On the sidebar, click on the Analytics tab to activate it.

Step 6: Under Custom, drag Reference Line from the Analytics tab and drop it onto the Table placeholder.

Step 7: Set the following for this Tableau reference lines:

- Scope as Entire Table
- Under Line, setValue, set the aggregation to Average of the SUM(Worldwide Gross Amount)
- Under Line Label, set it to Custom with Average <Value> as text; note you can use the > button to insert values
- Under Formatting, set Line to a thick, dark red line
- Under Formatting, set Fill Above: to None
- In Under Formatting, set Fill Below: to None:

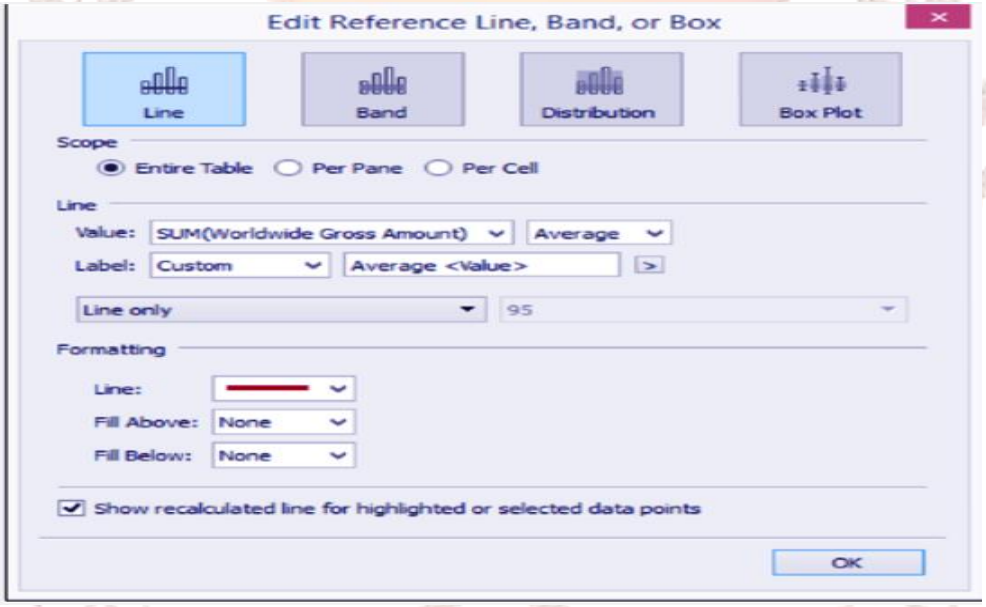


Fig 9.13 Reference lines and forecast

Step 8: Click OK when done.

Step 9: Under Custom, drag Reference Line again from the Analytics tab and this time drop it onto Pane.

SELF-ASSESSMENT QUESTIONS – 4

7. By definition, Tableau displays measures over time as a _____.
8. Tableau File Extension is_____

ANSWERS

7. Time
8. twbx

7. CREATING CALCULATED FIELDS

Step 1: Create the calculated field

1. In a worksheet in Tableau, select **Analysis > Create Calculated Field.**
2. In the Calculation Editor that opens, give the calculated field a name.

In this example, the calculated field is called Profit Ratio.

Step 2: Enter a formula

1. In the Calculation Editor, enter a formula.

This example uses the following formula:

SUM([Profit])/SUM([Sales])

Formulas use a combination of functions, fields, and operators. When finished, click **OK.**

The new calculated field is added to the Data pane. If the new field computes quantitative data, it is added to Measures. If it computes qualitative data, it is added to Dimensions.

You are now ready to use the calculated field in the view.

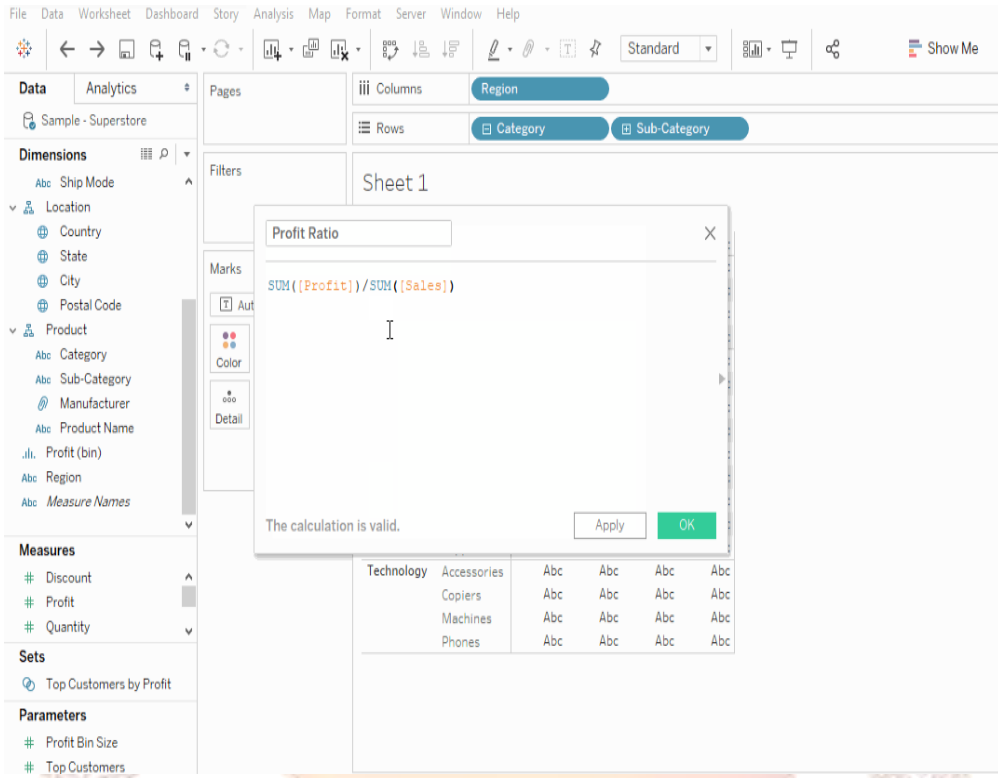


Fig 9.14 Creating calculated fields

SELF-ASSESSMENT QUESTIONS – 5

9. The icon associated with the field that has been grouped is a _____.
10. Tableau was introduced in the year of_____

ANSWERS

9. paper clip
10. 2003

8. TABLE CALCULATIONS

Quick table calculations allow you to quickly apply a common table calculation to your visualization using the most typical settings for that calculation type. This article demonstrates how to apply a quick table calculation to a visualization using an example.

The following quick table calculations are available in Tableau for you to use:

- Running total
- Difference
- Percent difference
- Percent of total
- Rank
- Percentile
- Moving average
- YTD total
- Compound growth rate
- Year of year growth
- YTD growth

Step 1: Open Tableau Desktop and connect to the **Sample-Superstore** data source, which comes with Tableau.

Step 2: Navigate to a new worksheet.

Step 3: From the **Data** pane, under Dimensions, drag **Order Date** to the **Columns** shelf.

Step 4: From the **Data** pane, under Dimensions, drag **State** to the **Rows** shelf.

Step 5: From the **Data** pane, under Measures, drag **Sales** to **Text** on the Marks Card.

Step 6: From the **Data** pane, under Measures, drag **Profit** to **Color** on the Marks Card.

Step 7: On the Marks card, click the Mark Type drop-down and select **Square.**

State	2014	2015	2016	2017
Alabama	$6,139	$3,892	$7,651	$1,828
Arizona	$8,295	$9,611	$6,242	$11,134
Arkansas	$6,303	$444	$2,224	$2,708
California	$91,304	$88,444	$131,552	$146,388
Colorado	$6,502	$4,639	$10,667	$10,300
Connecticut	$2,794	$1,285	$3,998	$5,307
Delaware	$4,786	$6,190	$2,720	$13,755
District of Columbia		$2,670	$117	$78
Florida	$34,248	$15,177	$13,603	$26,445
Georgia	$4,540	$11,338	$14,058	$19,160
Idaho	$465	$1,500	$1,183	$1,234
Illinois	$16,203	$18,578	$21,034	$24,352
Indiana	$2,937	$6,640	$25,462	$18,516
Iowa	$1,191	$1,713	$959	$716
Kansas	$1,320	$170	$691	$733
Kentucky	$8,280	$7,674	$5,103	$15,535
Louisiana	$2,535	$918	$263	$5,502
Maine	$617	$106	$547	
Maryland	$1,701	$7,843	$4,710	$9,452
Massachusetts	$9,900	$6,626	$3,965	$8,143
Michigan	$6,172	$16,844	$27,420	$25,834
Minnesota	$15,883	$5,724	$1,527	$6,728
Mississippi	$2,055	$302	$5,417	$2,997
Missouri	$1,648	$6,985	$4,221	$9,351
Montana	$424	$936		$4,229

Fig 9.15 Creating table calculations

SELF-ASSESSMENT QUESTIONS – 6

11. Sets can be created on Measures. _____
12. For creating variable size bins we use _____

ANSWERS

11. True
12. Calculated Fields

TERMINAL QUESTIONS

1. **What is a Calculated Field, and How Will You Create One?**

A calculated field is used to create new (modified) fields from existing data in the data source. It can be used to create more robust visualizations and doesn't affect the original dataset.

For example, let's calculate the "average delay to ship."

The data set considered here has information regarding order date and ship date for four different regions. To create a calculated field:

1. Go to Analysis and select Create Calculated Field.
2. A calculation editor pops up on the screen. Provide a name to the calculated field: Shipping Delay.
3. Enter the formula: DATEDIFF ('day', [Order Date], [Ship Date])
4. Click on Ok.
5. Bring Shipping Delay to the view.
6. Repeat steps 1 to 5 to create a new calculated field 'Average Shipping Delay' using the formula: AVG (DATEDIFF ('day,' [Order Date], [Ship Date]))

2. Is There a Difference Between Sets and Groups in Tableau?

A Tableau group is one dimensional, used to create a higher level category by using lower-level category members. Tableau sets can have conditions and can be grouped across multiple dimensions/measures.

Example: Sub-category can be grouped by category.

Top Sales and profit can be clubbed together for different categories by creating a set:

1. Continuing with the above example of Sets, select the Bottom Customers set where customer names are arranged based on profit.
2. Go to the 'Groups' tab and select the top five entries from the list.
3. Right-click and select create a group option.
4. Similarly, select the bottom five entries and create their group. Hide all the other entries.

3. What is a Parameter in Tableau? Give an Example.

A parameter is a dynamic value that a customer could select, and you can use it to replace constant values in calculations, filters, and reference lines.

4. How Can You Schedule a Workbook in Tableau after Publishing It?

1. When you're signed in to Tableau Server, go to Content > data sources or Content > Workbooks, depending on the type of content you want to refresh.
2. Select the checkbox for the data source or workbook you want to refresh, and then select Actions > Extract Refresh.
3. In the Refresh Extracts dialog, select Schedule a Refresh, and complete the following steps:

1. Select the schedule you want.
2. If available, specify whether you want a full or incremental refresh.

4. What Are the Different Types of Tableau?

The dif ferent types of Tableau are Desktop, Prep, Online, and Server.

5. How do I make sets in tableau?

Sets are custom fields that let you compare and ask questions about a subset of data. To make a set on a dimension, right-click on a size in the data pane and choose to create > set. On the "General" tab, select the fields used to figure out the set. On the conditions tab, you can set the conditions for making a set. On the top tab, you can also choose the top N members of the dataset based on any field. When a set is made, the measure is split into two parts, "in" and "out" of the set, based on the user's conditions.

6. How do I use groups in calculated fields in Tableau?

You can make a group by right-clicking on a field in the data pane and choosing "Create" > "Group." Then, you can select the fields you want to group under the "General" tab and set the criteria for grouping under the "Conditions" tab. Then, right-click on this group and choose "create," "set," and "create a calculated field." You can then use this group as a set in this calculated field.

7. What does "view in tableau" mean?

The term "view" refers to how data from a source is shown in a worksheet. A view can be anything. It could be a plot, a chart, a graph, or even a table. Then, all of these points of view are put together on a dashboard to make a single story and show how they all fit together

9. GLOSSARY

Let us have an overview of the important terms mentioned in the unit:

Group: A group is used to combine members in a field.

SETS: Sets are custom fields that define a subset of data based on some conditions.

10. CASELET

Lenovo designs develop, manufactures and market its product like-PC, laptop, tablet, mobile phones, servers, etc. Today, in 160 countries Lenovo expanded its empire.

The challenge

Creating reports in Excel was tiresome and required a team of 8 to 10 people for adoption to other divisions and regions. In addition to this, the analytics team spent six to seven hours creating one weekly report so, imagine what an overwhelming task it was to create 30 reports or more.

With Tableau, time spent on creating reports is much lesser than creating reports manually. Teams were able to deliver reports much faster, sometimes even reporting on daily or hourly basis.

The time saved is used in carrying out analysis and drawing insights from the information.

Implementation

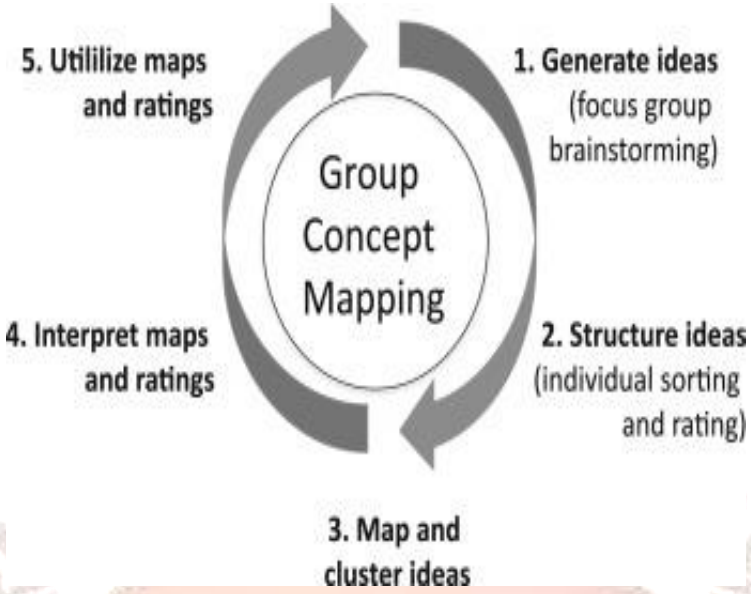
Initially, Lenovo deployed an eight-core instance of Tableau Server which they quickly scaled up to a 16-core server. The executives introduced Tableau in Lenovo India as a means to analyze and govern data on a selected set of business use cases and scenarios to help in decision making. But sooner than they realized, Tableau became an integral part of the company's functioning. The company experienced a cultural shift where the approach to business and growth was more data-centric and data-driven.

The change

- Lenovo India's BI Analytics & Visualization team created an interactive and flexible Tableau sales dashboard for departments to use it for ad-hoc analysis and reporting.

- Lenovo has over 55,000 employees and a customer base spread across 160 plus countries. More than 10,000 users access Tableau dashboards.
- Tableau has increased Lenovo's efficiency by 95% with approximately 3000 users are using it in about 28 countries by now.
- Lenovo's e-commerce team was able to analyze customer engagement patterns to improve brand perception and increase revenues.
- The human resource department converted 100 static reports into dynamic and interactive Tableau dashboards which gave users and analysts a new perspective into solving matters.
- The team is easily able to connect to data sources like Amazon Web Services and Hortonworks Hadoop Hive. Along with this, a wholesome analysis is possible through a dashboard where data is integrated from more than 30 data sources such as social media, customer surveys, retailer websites, online shopping sites, etc.
- Lenovo supports self-service analytics where every user can conduct an individual analysis on the set of data concerning their domain of activity and suiting their site roles. Every Tableau user has identification credentials stored in the local identity store of Tableau using which they can access and work on Tableau dashboards using single sign-on process.
- Lenovo experienced lucrative growth in e-commerce by using Tableau to analyze customer experience by fetching data from Lenovo's unified customer intelligence platform; LUCI Sky.

References and Suggested Reading

1. Joshua N. Milligan, Learning Tableau 2022: Create effective data visualizations, build interactive visual analytics, and improve your data storytelling capabilities, 5th Edition 5th ed. Edition
2. Visual Analytics with Tableau, Paperback, 31 May 2019.

11. CONCEPTUAL MAP